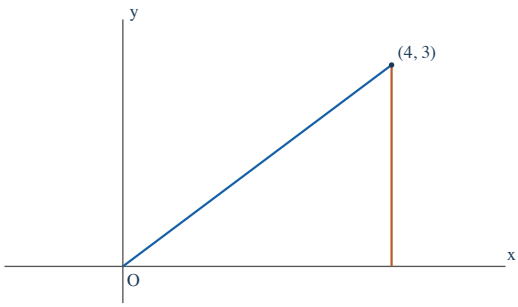
# Polar Coordinates

## What the devil are polar coordinates? 2

A polar point $(r, \theta)$ is a distance $r$ from the pole, at angle $\theta$ anticlockwise from the initial line. Angles are in radians.

Convert the Cartesian point $(4, 3)$ to polar form.




Answer

## Converting to/from polar coordinates 3

### Converting coordinates

$x = r\cos\theta,\ y = r\sin\theta,\ r^2 = x^2 + y^2$. Choose the angle from the correct quadrant.

Convert these Cartesian coordinates to polar form:
$(0, 2),\ (-3, 0),\ (1, 1),\ (-5, 12),\ (3\sqrt{3}, -3)$.

Polar coordinates

## Polar to Cartesian equations 4

Convert to Cartesian form:

a. $r = 5$

b. $r = 2 + \cos 2\theta$

c. $r^2 = \sin 2\theta$

a

b

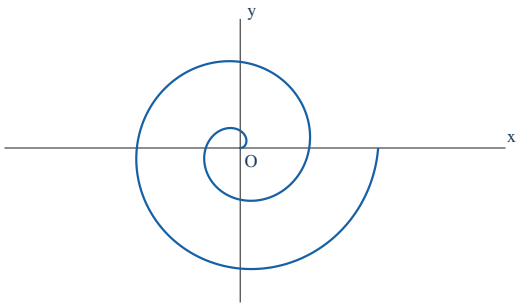
c

## Why polar form sometimes are preferable 5

**A spiral:** $r = \theta, \theta \geq 0$.

As the angle increases, the distance from the pole increases.

Its parametric form is $x = \theta \cos\theta,\ y = \theta \sin\theta$.




Why keep polar form?

## Cartesian to polar equations 6

Convert to polar form:

a. $y^2 = 4x$

b. $x^2 - y^2 = 5$

c. $\sqrt{3}y = x + 4$

a

b

c

2. Convert these polar coordinates to Cartesian coordinates:

a. $(6, \pi/6)$

b. $(6, -\pi/6)$

c. $(6, 3\pi/4)$

a

b

c



3a. Convert $r = 2$

3b. Convert $r = 3\sec\theta$

3c. Convert $r = 5\csc\theta$

a

b

c

3d. Convert $r = 4a\tan\theta\sec\theta$

3e. Convert $r = 2a\cos\theta$

3f. Convert $r = 3a\sin\theta$

d

e

f

4a. Convert $x^2 + y^2 = 16$

a

4b. Convert $xy = 4$

b

4c. Convert $(x^2 + y^2)^2 = 2xy$

c

4d. Convert $x^2 + y^2 - 2x = 0$

d

4e. Convert $(x + y)^2 = 4$

e

4f. Convert $x - y = 3$

f



5. Prove the distance formula for polar points $P = (r_1, \theta_1)$, $Q = (r_2, \theta_2)$.

Answer

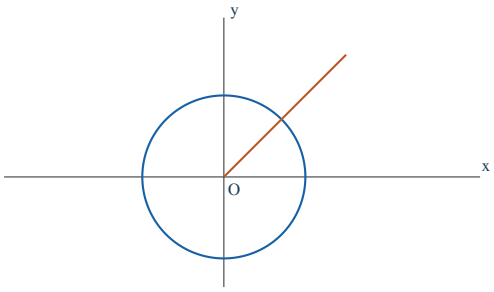
**Recognising polar curves**

$r = a$: circle, centre the pole, radius $a$.

$\theta = \alpha$, $r \geq 0$: a ray from the pole.

$r = a\theta$, $\theta \geq 0$: an Archimedean spiral.

Plot points and mark the direction of increasing angle.




## Sketching using tables of values 12

Sketch $r = \sin 4\theta$, using $r \geq 0$.

Start with $0 \leq \theta \leq \pi/4$.

Plot the loop

Sketch

## Sketching using tables of values 13

Sketch the cardioid $r = a(1 + \cos\theta)$, $a > 0$.

Key points

Sketch

## More sketchies 14

Sketch:

a. $r = a\sin 3\theta$, $r \geq 0$.

b. $r^2 = a^2 \cos 2\theta$.

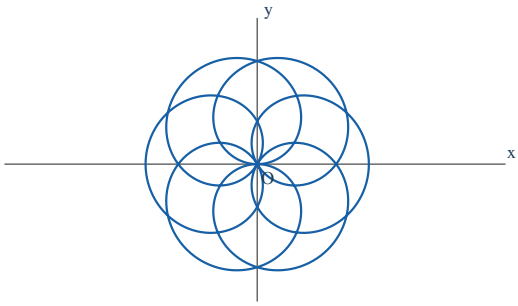
Answer

Sketch

**Exploration**

Plot $r = \cos(3\theta/4)$, allowing signed radii, for $0 \le \theta \le 8\pi$.

A negative radius is plotted in the opposite direction. This extension is separate from our usual $r \ge 0$ convention.



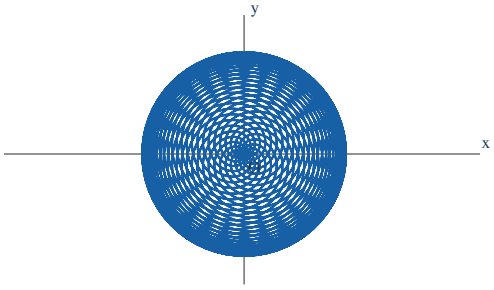

Observation

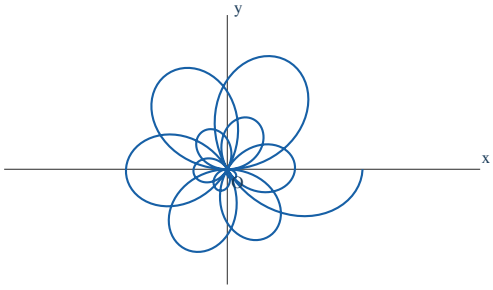


**More polar patterns**

Compare $r = \cos(13\theta/99)$ with $r = \theta\cos(3\theta/2)$, allowing signed radii.

The first produces a closed multi-loop pattern; the second expands outward as the angle increases.

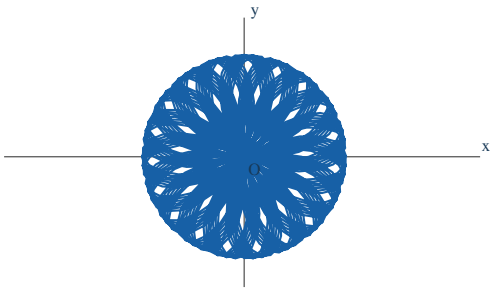


**An irrational frequency**

Plot $r = \cos(e\theta)$ for $0 \leq \theta \leq 50\pi$, allowing signed radii.

Since $e$ is irrational, there is no common angular period that returns the full parametrisation to its starting state. The pattern does not close exactly.






**Limaçons**

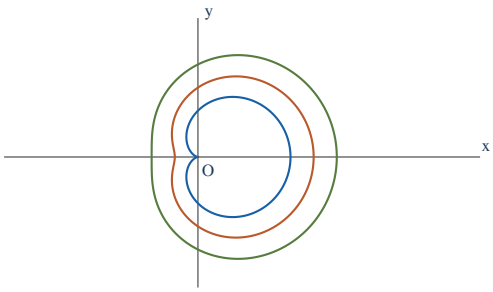
$r = a(p + q\cos\theta)$, with $a, q > 0, p \geq q$.

$p = q$: cardioid with a cusp.

$q < p < 2q$: dimpled limaçon.

$p \geq 2q$: convex limaçon.

The curve is symmetric in the initial line.



## Egg vs Dimple 19

Sketch and compare:

a. $r = a(5 + 2\cos\theta)$

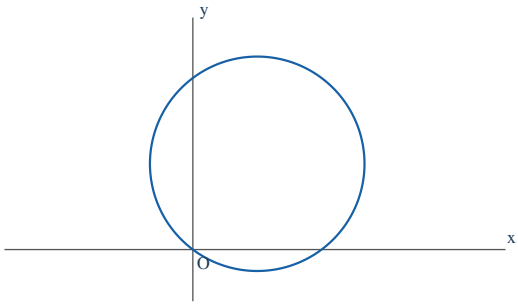
b. $r = a(3 + 2\cos\theta)$

Answer

Sketch

## Final Example 20

The locus $|z - 3 - 4i| = 5$ is a circle. Find its polar equation.




Answer

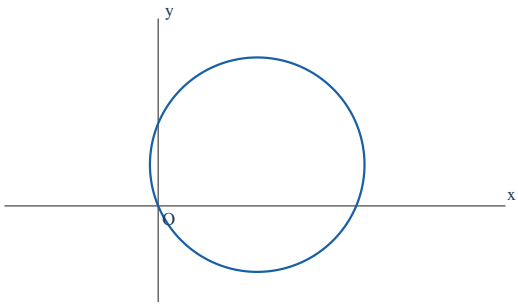
2. Sketch $r = k\sec(\pi/4 - \theta)$, $k > 0$, giving the axis intercepts.

Answer



3a. Sketch $|z - 12 - 5i| = 13$ on an Argand diagram.

b. Show its polar equation is $r = 24\cos\theta + 10\sin\theta$.




a

b

**Curve summary**

$r = a$: circle at the pole.

$\theta = \alpha$: ray at angle $\alpha$.

$r = 2a\cos\theta$: circle centred at $(a, 0)$, radius $a$.

$r = k\theta$: spiral.

Always state the angular interval and whether signed radii are permitted.



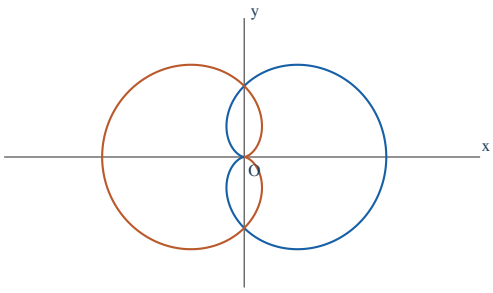
**Symmetry and orientation**

$a(1 + \cos\theta)$: cardioid facing right.

$a(1 - \cos\theta)$: cardioid facing left.

Replacing cosine by sine rotates the orientation.

$a(3 + 2\cos\theta)$: a dimpled limaçon, with minimum radius $a$.



**More standard forms**

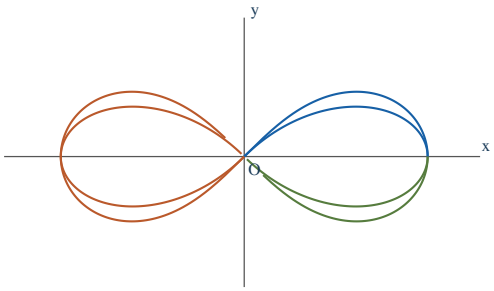
$r = a\cos 2\theta$, $r \geq 0$: two positive-radius petals.

$r^2 = a^2 \cos 2\theta$: lemniscate.

$r = p\sec(\alpha - \theta)$: straight line

$x\cos\alpha + y\sin\alpha = p$.

In particular, $r = a\sec\theta$ gives $x = a$.






**From strips to sectors**

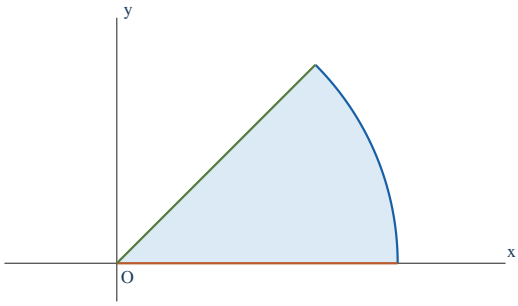
In Cartesian coordinates, area is accumulated from thin strips: $dA = y\,dx$.

A solid of revolution uses thin discs: $dV = \pi y^2 dx$.

For polar area, use thin sectors with angle $d\theta$.

## Area in polar coordinates

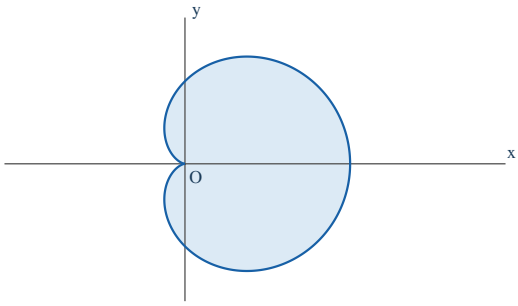
A sector of radius $r$ and angle $d\theta$ has area approximately $\frac{1}{2}r^2 d\theta$.




Formula



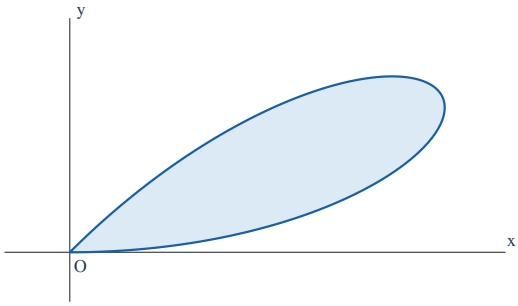
Find the area enclosed by $r = a(1 + \cos\theta)$.




Answer

## Another Example 29

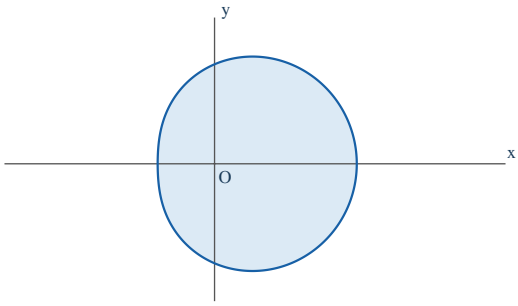
Find the area of one loop of $r = a \sin 4\theta$, $r \geq 0$.




Answer

## Test Your Understanding So Far 30

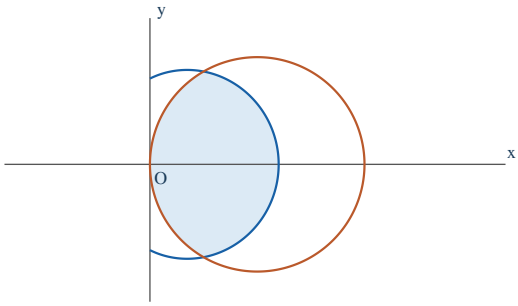
The area enclosed by $r = a + 3\cos\theta$, $a > 3$, is $107\pi/2$. Find $a$.




Answer

## Intersecting Areas 31

Find the area common to $r = 2 + \cos\theta$ and $r = 5\cos\theta$.

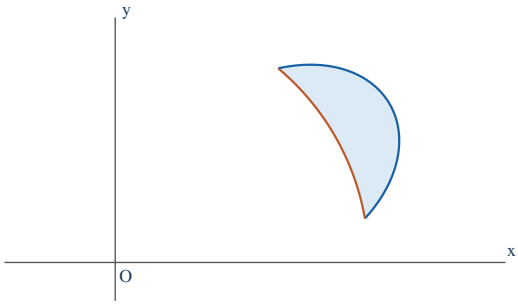



Intersections

Area

## Test Your Understanding 32

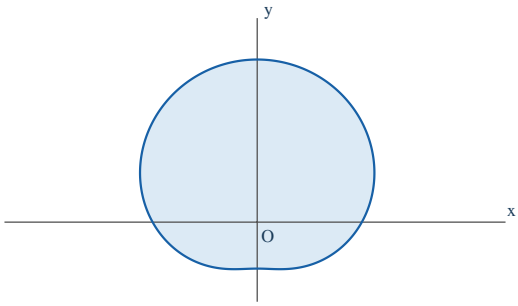
Find the area outside $r = 2$ but inside $r = \frac{3}{2} + \sin 3\theta$, for $0 \leq \theta \leq \pi/2$.




Intersections

Area

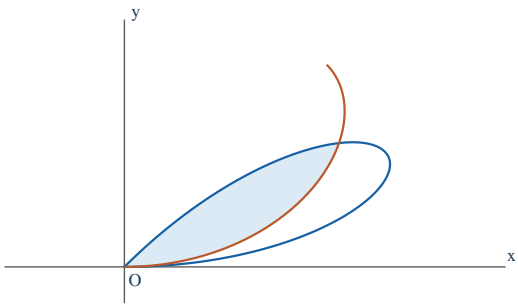
4. The area enclosed by $r = a + 5\sin\theta$, $a > 5$, is $187\pi/2$. Find $a$.




Answer



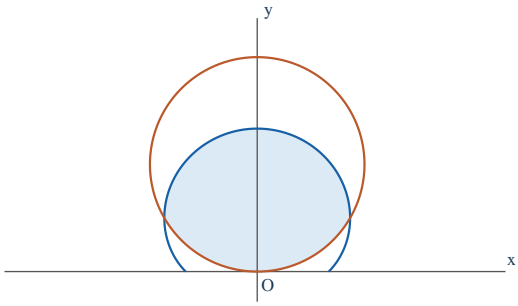
5. Find the finite area common to $r = a\sin 4\theta$ and $r = a\sin 2\theta$, with $0 \leq \theta \leq \pi/2$ and $r \geq 0$.




Intersections

Area

6. Find the area common to $r = 1 + \sin\theta$ and $r = 3\sin\theta$.

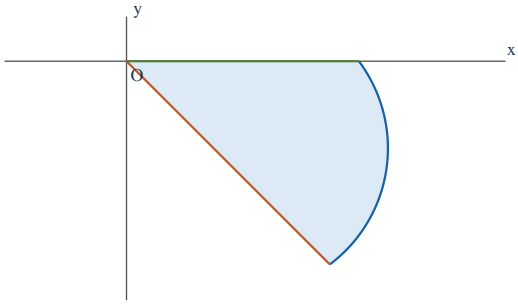



Intersections

Use symmetry



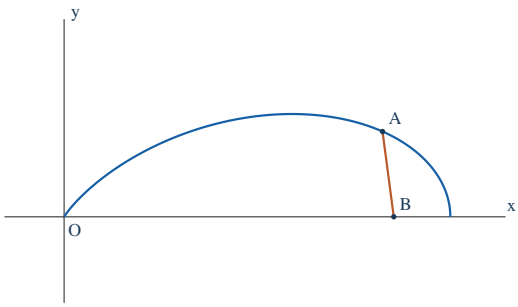
7. Sketch the region $-\pi/4 \leq \arg z \leq 0$, $|z - 4 + 3i| \leq 5$. Find its area to 3 significant figures. The circle has polar equation $r = 8\cos\theta - 6\sin\theta$.




Sketch

Area

9. On $r = 1 + \cos 3\theta$, $0 \leq \theta \leq \pi/3$, point A has radius $(2 + \sqrt{2})/2$. Point B is on the initial line with the same radius. Find the area between arc A to $(2, 0)$, AB and the initial line, to 3 significant figures.

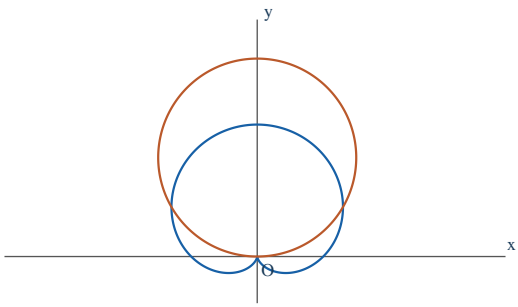



Sector

Subtract triangle



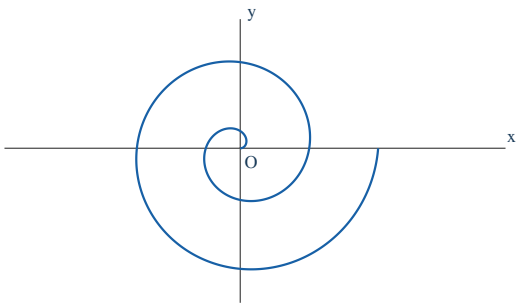
10. Find the area inside the cardioid $r = 1 + \sin\theta$ and outside the circle $r = 3\sin\theta$, to 2 decimal places.




Answer

## Challenge 5C - Core Pure Year 2 Pages 111-112 39

Challenge. A shell is modelled by $r = k\theta$, $0 \leq \theta \leq 4\pi$. The width along the initial line, from the point at $3\pi$ to the point at $4\pi$, is 3 cm. Find $k$ and the area inside the outer turn.




Scale

Area

## Tangents and Normals 40

### Tangents to polar curves

$x = r\cos\theta, \qquad y = r\sin\theta$

Differentiate

## Example 41

Find the regular points where $r = a(1 + \cos\theta)$ has horizontal tangents.

Answer

Sketch

## Test Your Understanding So Far 42

For $r = 1 + 2\cos\theta$, $0 \leq \theta \leq \pi/2$, find the radius at the point with a horizontal tangent.

Answer

Sketch

## Example 43

Find horizontal tangents to $r = a \sin 2\theta$, $0 \leq \theta \leq \pi/2$.

Answer

Sketch

## Example 44

Find vertical tangents to $r = a \sin 2\theta$, $0 \leq \theta \leq \pi/2$.

Answer

Sketch

## Proof of dimple vs egg 45

Explain why $r = a(p + q\cos\theta)$, $p > q > 0$, has a dimple when $p < 2q$.

Answer

## Exercise 5D - Core Pure Year 2 Pages 115-116 46

4. Find the polar coordinates on $r = a(7 + 2\cos\theta)$ where the tangents are parallel to the initial line.

Answer

Sketch

5. Find the tangent lines to $r = 2 + \cos\theta$ that are perpendicular to the initial line.

Answer

Sketch



6. Find the point on $r = a(1 + \tan\theta)$, $0 \leq \theta < \pi/2$, with a tangent perpendicular to the initial line.

Answer

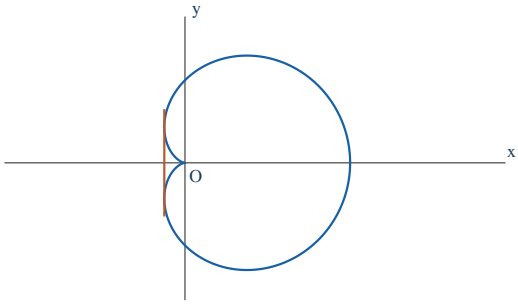
7. A point A on $r = 1 + 3\cos\theta$, $0 \leq \theta \leq \pi/2$, has a horizontal tangent. Find the exact length OA.

Answer

Sketch



8. For the cardioid $r = 2(1 + \cos\theta)$, find the area between the curve and its left-hand vertical common tangent, to 3 significant figures.




Tangencies

Area